Module 6 Lecture - Surveys & Observational Design

Research Methods in Psychology

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1 Overview and Introduction

1.1 Learning Objective(s)

- Explain how carefully prepared questions improve the construct validity of a poll or survey.
- Describe how researchers can make observations with good construct validity.
- Be able to apply the concepts of chapter 5 measurement reliability and validity to self-report and observational measures/tools
- Understand the nuance and drawbacks of certain strategies in creating measures, as well as common pitfalls in writing questions and structuring answers
- Have an idea of strategies that can be employed to minimize problems in surveys and observations.

1.2 Overview

- Observational design is when we don't attempt to _____ or manipulate any variables, we just take measurements of individuals in their natural disposition, i.e., just lots of **measured variables**, no **manipulated variables**
- Often, we use these _____ for more macro-level studies that look for big trends across people, and they tend to be most effective in assessing _____ claims (though can be appropriate for association and causal claims as well)
- Polls and observational designs are everywhere:
  - _____ polls (technically even the election itself, is effectively a survey)
  - Public _____ polls on certain topics
  - Interest in a certain commercial product

2 Construct Validity in Survey Designs and Self-reports

2.1 Overview

- Surveys, polls, and observational designs mean the same thing, and describe a _____ by which data is gathered from a certain sample

via a self-report

- This design has plenty of _____, such as being done via the mail, email, phone, advertisements, etc. - which change the construct validity of the measurements done

2.2 Question Formats

- Question types range from _____ to most restrictive, with different types having varying _____ to use.
- Open-ended questions are those that allow respondents to reply in _____ and paragraphs not necessarily bound to one format. While capturing the "most" _____, these questions are difficult to transform into quantitative findings - in fact, some researchers would say they should not be transformed at all
  - This is the most common type of _____ in *qualitative research*, where the goal is to often capture the full, anecdotal experience of participants.
  - Note: In this class, we will be almost entirely focused on *quantitative research*, which is that which uses _____ analysis and operationalized experiences to numbers.
- Forced-choice questions are those that _____ respondents to only responding to a question or prompt is a specified number of ways.
  - For example, any b58fc729-690b-4000-b19f-365a4093b2ff-7B7B3C20626C616E6B206D756C7469706C or true-false assessment is forced choice.
  - Think about how forced-choice naturally _____ participants in sharing the full breadth of their experience.
- Likert-scale questions is ordinal scale question that asks a participant to respond to a statement or questions with 5 _____ answers:
  - Strongly agree
  - Agree
  - Neither agree nor disagree
  - Disagree
  - Strongly Disagree
- A question that is similar in structure but has _____ or less options than those 5 should be referred to as a Likert-type scale question
- Semantic differential is a format that asks a respondent to respond to a

prompt with a "rating" between two _____ or adjectives. Ex. This could be a "star" system for rating satisfaction.

- Question formats do not inherently _____ or add to construct validity, *however* they will have an impact on the type of analysis one can use. Likert, Likert-type, and Semantic _____ will all produce ordinal data. Forced choice is likely to be _____. And open-ended is a whole other thing entirely…
  - Be mindful of how you will perform analysis *before making a survey or self report tool!*

2.3 "Good" Questions

- Question writing can have a large impact on the _____ and construct validity of a question - writing good questions is often time-consuming and intense!
- While there are many ways for a question to go "wrong", There are a few pitfalls that are especially _____ and dangerous…

2.3.1 Leading Questions

- Leading questions are when a question is worded or designed in such a way that is likely to _____ respondents towards a certain answer or outcome. This can be accidental, or intentional.
- In general, avoid using terms that are _____ in nature - like "awful", "bad", "dangerous", etc. If your questions is likely to play up the emotion of your participant, it is possibly a leading question.

2.3.2 Double-barreled Questions

- Double-barreled questions when a question is actually two questions _____ into one. The problem is that this might cause confusion in capturing the real opinions or feelings of an individual.
- A good way to _____ this mistake is if you see an "and" anywhere in the question - tread cautiously

2.3.3 Negative Wording

- Negative wording is when a question is worded in such a manner that confounds the _____ of a question.
  - Ex. "Do you *not* agree with…"
- Not only can this be difficult to properly analyze in a study, it can also be generally very _____ to participants and produce inaccurate responses.
- Wherever possible, avoid "not", "nor", "neither" and other negative words in surveys. Sometimes, these sorts of issues can be rectified by _____ a question up, similar to procedures for double-barreled questions.
- If a question is split up to be both a negative and positive version, one can use a correlation _____ and Cronbach's α to ensure that same-direction questions correlate well with one another (and if they don't - we have a problem)

2.3.4 Question Order

- This is a complicated issue and can be hard to fully prevent problems with. Identifying this issue is sometimes best investigated through a pilot study, which occurs _____ the primary study, and is usually meant to ensure that a measure is sound before rolling out to the "real sample".
- To catch a question _____ effect, you may use two different versions of the measure, with a different order of questions, and give those versions to two different groups. Then _____ the answers of the groups to see if they are similar or not. If they differ, there is something in the order affecting the scores.
- "Solving" this issue often involves a good, _____ understanding of the literature and some _____ as to why participants may react a certain way to delicate questions.

2.4 Getting Accurate Responses

- Surveys are, effectively, always a b58fc729-690b-4000-b19f-365a4093b2ff-7B7B3C20626C616E6B2073656C66203E7D7D–report - they require an introspection on the part of the respondent. In a lot of research, we must trust

and place _____ in the ability of a person to report on their internal experiences.

- However, for a variety of reasons, we must be cautious of certain response sets, which are when a participant follows a _____ of responses which may not be particularly informative. In the worst case, a response set represents _____ responding by the participant, which confounds results.
- Response sets generally occur more _____ in Likert or Likert-type scales

2.4.1 Acquiescence / Yea-saying

- Acquiescence / Yea-saying occurs when a participant carelessly selects the most positive _____ (e.g., Strongly Agree) or yes throughout a measure
- When a respondent is yea-saying, it makes it incredibly difficult to discern whether that _____ represents their opinions/disposition.
- We may try to use reverse-coding questions to detect this - Ex:
  - "I feel happy most of the time"
  - "I feel sad most of the time"

2.4.2 Fence Sitting

- Fence-sitting is when a person keeps choosing the _____ or neutral option
- This can be resolved by _____ the middle option, but this also limits the inclusiveness of the question. We may also choose to use a forced-choice self-report to force choosing between categories.

2.4.3 Socially Desirable Responding / Faking Good

- This is when one takes a certain response trend where they intentionally try to respond in a way that seems _____ to most people, maybe due to embarrassment or shame
  - Ex., On a personality scale I answer in an overly-altruistic manner

- We may also be concerned with faking bad/malingering, which is especially true in _____ or neuropsychological settings
- We can help limit these sets by including special _____ questions that would be absurd for *anyone* to be *that* good or bad. We can also ensure participants know results are _____. Finally, we may also get converging evidence from collateral reports as well.

2.5 Other Confounds in Surveys

- We may often run into other unexpected problems with our self reports and surveys in general related to people's ability to _____ accurately
- One issue is that people sometimes may choose options intuitively, and may not know *why* they made a choice. There may be a number of underlying cognitive reasons for _____ a certain way, even if people don't know it!
- Accuracy of _____, especially, can be a tricky subject and when we ask for people to report on more distant memories, we may get responses that may not be quite accurate. Just because someone reports _____ in their memory, also does not necessarily mean that it is accurate. The best way we can navigate this issue is try to get additional information or sources.

3 Construct Validity in Behavioral Observations

3.1 Overview

- Observational measures are _____ to frequency, association, and causal claims. Observational measures also give a sense of "objectiveness", as they are not _____ on the ability of a person to accurately introspect and report feelings.
- Think about the historical movement away from _____ and psychodynamicism to behaviorism - just like those early behaviorists, some see observational data as the "superior" type of measure.
- However, just like with self-reports, we must be _____ of the construct validity of these observations.

3.2 Claims on Observational Data

- Observational data is all about behaviors and what we can tangibly sense with our _____ about what a participant is doing. Most of the time, we are looking at what a person is doing.
- For some cases, it might be more *face* _____ to just measure a behavior to tie back to a construct

3.3 Reliability and Validity in Observation Data

- Construct validity in observational data can be confounded by 3 types of biases: _____ bias, observer effects, and reactivity.

3.3.1 Observer Bias

- (Observer bias) Depending on the circumstances, observers may be biased to "see" a certain _____ in participants due to some preconceived notions or beliefs
- This is why it is important to have observers _____ to the study they may be observing and to have rigorous training that helps ensure a smooth and consistent process (double-checked with reliability analysis! - but what type…)
  - It is common to use codebooks to clearly highlight how a variable or behavior is to be _____ for a particular study.
- Blinded studies may also be called _____ studies.

3.3.2 Observer Effects

- Observer effects are when participants act un-natural in response to the _____ or perspectives of the researchers. Participants may readily act a certain way to appear "good" or they may be _____ careful or stiff or modify actions as they watch how experiments react.
- Participants may also be *unintentionally* reacting to the behaviors of the observers as well!
- The same solutions can help, especially blinding the _____, because they won't be unconsciously swayed by their knowledge.

3.3.3 Reactivity

- Reactivity is somewhat similar to Observer effects, but hinges on the mere _____ of the observer causing behavior changes in the participant.
- Three solutions to minimize reactive behaviors
  - Find some way to be "unobtrusive" in observation, whether viewing from afar or outside the _____ of the subject
  - Allow a subject to _____ to having you there and let any initial reactivity die out
  - Measure _____ results of the behavior once the subject themself is gone - but this is still observational

3.3.4 Ethics in Observation

- Like any procedures planned for research, observation techniques and tools need to be approached in an ethical and sensible manner, with informed consent/_____ telling participants how and why their data is used
- Some deception, such as watching through a one-way mirror may be _____ (if allowed by the IRB), as long as it has good, scientific rationale.

4 Conclusion

4.1 Recap

- This unit covered the process of conducting observational studies and being aware of the problems that may arise in these studies
- We discussed the various solutions we might consider to help keep our construct validity strong when doing observational designs
- We also reviewed on how different types of tools (self-report, observational) require different approaches to maximize validity

4.2 Review Questions

? Which of the following is not one of the way we discussed can cause problems in participants understanding of questions?

- (●) A) Leading questions
- ( ) B) Double-barreled questions
- ( ) C) Negative wording
- ( ) D) Reverse coding

Explanation:

Discuss: In the last question, I used the word 'not' in the text, what would we call this and what problems might it create?

? Which of the following is a problem relevant to self-report surveys?

- (●) A) Reactivity
- ( ) B) Observer bias
- ( ) C) Fence sitting
- ( ) D) Observer effect

Explanation:

Discuss: If self-report and observational measures are two types of measures we've talked about previously, what is the third type of measure not discussed in this lecture?

A person is asked to rate themselves on a 5-point scale between 'sad' and 'happy', this is:

- (●) A) Semantic differential
- ( ) B) Likert-scale
- ( ) C) Likert-type
- ( ) D) Open-ended

Explanation:

Discuss: What do you think is a weakness of open-ended questions on surveys?

Discuss: Consider this question: 'Are you impacted by any recent stressful personal and/or world events right now, yes or no?' Do you see any issues with this question in its current form?

Which of the following can help with both observer bias and observer effects?

- (●) A) Blinding
- ( ) B) Codebooks
- ( ) C) Making observers unobtrusive or unseen
- ( ) D) None of the above

Explanation:

Key Terms

A

Acquiescence / Yea-saying Response set where a person answers 'yes' or 'agree' or 'strongly agree' throughout a survey 6

B

Blinded studies Studies conducted so that the participants, observers, or both, are unaware of the research question / hypothesis 8

C

codebooks Detailed guides to be used by observers in observational studies, to reduce observer bias 8

D

Double-barreled questions A question written in such a way that it has two questions included within it 4

F

faking bad/malingering When a person purposefully rates or behaves worse for secondary gain 7
Fence-sitting When a person keep choosing the most neutral option available 6
Forced-choice questions Questions asked in a way so that respondents may only select from a limited number of answers 3

L

Leading questions Questions written in a way to bias a participant towards a certain outcome 4
Likert-scale questions Those asked in a disagree-agree scale, with 5 options 3
Likert-type scale question Similar to Likert, but with less than or more than 5 options 3

N

Negative wording When a question is worded with a not or neither that switches the 'direction' of the question 5

O

Observational design A type of research design that focuses on gather evidence by only measuring variables, without manipulation 2
Observer bias When observers ratings or scores of participants are biased by the raters preconceived beliefs 8
Observer effects When the observer unintentionally bias participants towards a certain reaction or behavior 8
Open-ended questions Questions asked in a way that allow a respondent to respond in full sentences or narrative 3

P

pilot study A small study run before the main one, in order to test whether the study 'works' as planned on a smaller group 5
polls Another name for observational designs when gathering self-reported data 2

R

Reactivity When a participant is biased by the mere presence of a researcher or observer 9
response sets When a participants answers in certain, unrealistic trend 6
reverse-coding A process by which we include questions that are 'scored' in a different direction, usually to try to detect acquiescence 6

S

Semantic differential Questions asked in such a manner that ask a person to rate between two adjectives 3

Surveys Another name for observational designs when gather self reported data 2

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms